$$K_a = \frac{\Lambda_m^2 \, C}{\Lambda_m^{\circ 2} \frac{(\Lambda_m^{\circ} - \Lambda_m)}{\Lambda_m^{\circ}}}$$

$$\Rightarrow K_a = \frac{\Lambda_m^2 C}{\Lambda_m^{\circ} (\Lambda_m^{\circ} - \Lambda_m)} \qquad .....(9.16)$$

3. Calculation of solubility of sparingly soluble salts

Substances like AgCl, $PbSO_4$ etc., are sparingly soluble in water. The solubility product of such substances can be determined using conductivity measurements.

Let us consider AgCl as an example

$$AgCl\,(s) \rightleftharpoons Ag^+ + Cl^-$$

$$K_{sp} = [Ag^+][Cl^-]$$

Let the concentration of $[Ag^+]$ be 'C' $molL^{-1}$.

As per the stoichiometry, if $[Ag^+] = C$, then $[Cl^-]$ also equal to 'C' mol L^{-1}.

$$K_{sp} = C.C$$

$$\Rightarrow K_{sp} = C^2$$

We know that the concentration (in mol dm^{-3}) is related to the molar and specific conductance by the following expressions

$$\Lambda^0 = \frac{\kappa \times 10^{-3}}{C \text{ (in mol } L^{-1})}$$

(or)

$$C = \frac{\kappa \times 10^{-3}}{\Lambda^0}$$

Substitute the concentration value in the relation $K_{sp} = C^2$

$$K_{sp} = \left(\frac{\kappa \times 10^{-3}}{\Lambda^{\circ}} \right)^2 \qquad .....(9.17)$$

9.3 Electrochemical Cell

Electrochemical cell is a device which converts chemical energy into electrical energy and vice versa. It consists of two separate electrodes which are in contact with an electrolyte solution. Electrochemical cells are mainly classified into the following two types.

1. **Galvanic Cell (Voltaic cell)** : It is a device in which a spontaneous chemical reaction generates an electric current i.e., it converts chemical energy into electrical energy. It is commonly known as a battery.
2. **Electrolytic cell** : It is a device in which an electric current from an external source drives a nonspontaneous reaction i.e., it converts electrical energy into chemical energy.

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